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SUPERSEDING
FA-PD-MI-2597, Rev. 0

MILITARY SPECIFICATION

PROJECTILE, 90MM, TP-T, M353
METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Projectile, 90MM, TP-T, M353 Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military
MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

Federal
Federal Test Method Standard No. 151 - Metals; Test Methods

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-639 - Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167 - Ammunition Data Cards
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1315

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DRAWINGS

Munitions Command

| | |
|---|---|
| F-10522528 | - Projectile for Cartridge, 90MM, TP-T, M353, Metal Parts Assembly |
| F-7548604 | - Pallet for Fixed and Semi-Fixed Projectiles; Assembly, Details, Packing and Marking |
| IEL-10522528 | - Inspection Equipment List for Projectile, For Cartridge, 90MM, TP-T, M353, Metal Parts Assembly |

PUBLICATIONS

Munitions Command
Frankford Arsenal
Operating Instructions for Use of Detroit Testing Machine for Band Seating Inspection

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements of the applicable drawings, referenced specifications and the following:

3.2 First article sample.- Requirements for submission of first article sample by the contractor shall be as specified in the contract.

3.3 Projectile body.

3.3.1 Metal defects.- The projectile body shall not contain cracks. Imperfections on non-metallic inclusions which are greater than two inches in length are not permissible.

3.3.2 Band seat.- Prior to application of rotating band blank, the band seat shall be clean and free of oil, grease, dirt, rust and other foreign matter. Shot or sand blasting shall not be used in cleaning the band seat after knurling.

3.3.3 Protective coating.- Protective coating shall be in accordance with the applicable drawings and specifications referenced thereon.



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3.4 Rotating band.

3.4.1 Blank.- The rotating band blank may be applied either hot or cold; if applied hot, the blank shall not be heated above 1600 degrees Fahrenheit preparatory to application, and scale formation shall be held to a minimum.

3.4.2 Mechanical properties.- After seating and subsequent removal from the projectile body, the rotating band shall have the following minimum mechanical properties:

| | | |
|---|---|---|
| Tensile strength | - | 32,000 pounds per square inch (psi) |
| Elongation | - | 15 percent for one inch of gage length |

3.4.3 Band seating.- The diametrical clearance between the rotating band and band seat shall not exceed .006 inch.

3.4.4 Protection.- The machined rotating band shall be carefully protected from being nicked, burred or otherwise damaged in subsequent handling and shipping.

3.5 Proving ground functional tests.- There shall be no evidence of separation or breakup of components when the metal parts assembly is tested in accordance with 4.4.5.1 and 4.4.5.2.

3.6 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects, non-ferrous.- Soundness of material shall be as required by the applicable specification.

3.6.2 Threads.- Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable component drawing.

3.6.3 Burr.- No part shall have a burr which might interfere with the assembly or function of the projectile, or which might be injurious to personnel during handling.

3.6.4 Foreign matter.- No part or assembly shall contain dirt, chips, grease, oil, rust, corrosion or other foreign matter.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.1.3 Inspection criteria.-

a. Successful passage of all inspections required by the drawings and specifications shall be considered as a condition precedent to performing the Proving Ground functional tests (see 3.5).

b. Acceptance or rejection of a lot by the Government shall be contingent on successful passage of the Proving Ground functional tests in 4.3.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 32 assemblies as follows:

25 Metal Parts Assemblies complete including paint
2 Metal Parts Assemblies complete except paint
5 sets of component parts, unassembled and unpainted, consisting of:
- 5 each bodies prior to banding
- 5 each rotating band blanks
- 5 each windshields

Twenty (20) of the above complete metal parts assemblies shall be forwarded to a designated Proving Ground for ballistic testing. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.



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4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all the requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of Product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105 or MIL-STD-1235, as applicable. An AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used. In performing visual examination for such defects as burr and foreign matter, the Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination.

4.3.2.1 Windshield, Drawing C10522531 covering a detail of Drawing F10522528.

| Defect | | Method of Inspection |
|---|---|---|
| CRITICAL | | |
| None defined | | |
| MAJOR | | |
| 101. | Pitch diameter of thread, max. | Gage |
| 102. | Minor diameter of thread, max. | Gage |
| 103. | Length of effective thread, min. | Gage |
| 104. | Thickness of wall, min. | Gage |
| 105. | Metal defective | Visual |



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4.3.2.1 Cont'd

| Defect | Method of Inspection |
|---|---|
| MINOR | |
| 201. Diameter at rear, outside | Gage |
| 202. Length from rear face to forward outside basic diameter | Gage |
| 203. Length from rear face to rear outside basic diameter | Gage |
| 204. Length of thread, max. | Gage |
| 205. Length, total | Gage |
| 206. Surface finish, improper | Visual (see 6.1) |
| 207. Radii or chamfers, missing or incorrect | Visual (see 6.1) |
| 208. Burr | Visual |
| 209. Foreign matter | Visual |

4.3.2.2 Body, Drawing F10522530 covering a detail of Drawing F10522528.

CRITICAL

None defined

| | |
|---|---|
| MAJOR | |
| 101. Diameter of rotating band seat | Gage |
| 102. Scoring or knurling missing from band seat | Visual |
| 103. Metal defective | Visual |
| MINOR | |
| 201. Location of rotating band seat | Gage |
| 202. Width of rotating band seat | Gage |
| 203. Surface finish, improper | Visual (see 6.1) |
| 204. Radii or chamfers, missing or incorrect | Visual (see 6.1) |
| 205. Burr | Visual |
| 206. Foreign matter | Visual |

4.3.2.3 Body Assembly, Drawing D10522529 covering a detail of Drawing F10522528.

CRITICAL

None defined

| | |
|---|---|
| MAJOR | |
| 101. Pitch diameter of windshield thread, min. | Gage |
| 102. Major diameter of windshield thread, min. | Gage |



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4.3.2.3 Cont'd

| Defect | Method of Inspection |
|---|---|
| MINOR | |
| 201. Diameter forward of windshield thread | Gage |
| 202. Diameter of windshield thread underdut | Gage |
| 203. Angle forward of windshield thread | Gage |
| 204. Angle forward of bourrelet, min. | Gage |
| 205. Length from shoulder to start of threads | Gage |
| 206. Length from rear face to shoulder | Gage |
| 207. Length of angle forward of bourrelet, min. | Gage |
| 208. Length from shoulder to angle forward of windshield thread | Gage |
| 209. Width of windshield thread undercut | Gage |
| 210. Surface finish, improper | Visual (see 6.1) |
| 211. Radii or chamfers, missing or incorrect | Visual (see 6.1) |
| 212. Burr | Visual |
| 213. Foreign matter | Visual |

4.3.2.4 Metal Parts Assembly, Drawing F10522528 (before Paint).

| | |
|---|---|
| CRITICAL | |
| None defined | |
| AMAJOR | |
| 101. Insecure assembly | Torque wrench (see 4.4) |
| 102. Concentricity of forward taper of windshield with bourrelet | Gage |
| 103. Concentricity of rear of windshield with bourrelet | Gage |
| 104. Concentricity of body with bourrelet | Gage |
| 105. Concentricity of rotating band with bourrelet | Gage |
| 106. Concentricity of tracer cavity with bourrelet | Gage |
| 107. Diameter of bourrelets, min. | Gage |
| 108. Diameter of rotating band | Gage |
| 109. Depth of crimp groove | Gage |
| 110. Length of effective tracer thread, min. | Gage |
| 111. Location of crimp groove | Gage |
| 112. Location of rotating band | Gage |
| 113. Radiusat rear of crimp groove | Gage |
| 114. Width of rotating band | Gage |
| 115. Gap between windshield and body | Gage |



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4.3.2.4 Cont'd

| | Defect | Method of Inspection |
|---|---|---|
| MINOR | | |
| 201. | Diameter of body | Gage |
| 202. | Diameter of rotating band relief groove | Gage |
| 203. | Depth to bottom of tracer cavity | Gage |
| 204. | Angle at rear of rotating band relief groove | Gage |
| 205. | Length total, max. | Gage |
| 206. | Surface finish, improper | Visual (see 6.1) |
| 207. | Radii or chamfers, missing or incorrect | Visual (see 6.1) |
| 208. | Burr | Visual |
| 209. | Foreign matter | Visual |

4.3.2.5 Metal Parts Assembly, Drawing F10522528 (after paint).

| | | |
|---|---|---|
| CRITICAL | | |
| | None defined | |
| MAJOR | | |
| 101. | Weight | Approved scale |
| 102. | Tracer cavity not functionally concentric with minor diameter of tracer thread | Gage |
| 103. | Pitch diameter of tracer thread, max. | Gage |
| 104. | Minor diameter of tracer thread, max. | Gage |
| 105. | Diameter of bourrelets, max. | Gage |
| 106. | Rotating band damaged | Visual |
| MINOR | | |
| 201. | Pitch diameter of tracer thread, min. | Gage |
| 202. | Diameter of tracer cavity | Gage |
| 203. | Protective coating inadequate or damaged | Visual |
| 204. | Paint on rotating band bearing surfaces | Visual |
| 205. | Paint on threads | Visual |
| 206. | Paint in tracer cavity | Visual |
| 207. | Marking missing, incorrect or illegible | Visual |
| 208. | Burr | Visual |
| 209. | Foreign matter | Visual |



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4.3.2.6 Plug and Disc Assembly, Drawing B8597489 covering a detail of Drawing F10522528.

| Defect | Method of Inspection |
|---|---|
| **CRITICAL** | |
| None defined | |
| **MAJOR** | |
| 101. Inside diameter not functionally concentric with major diameter of thread | Gage |
| 102. Pitch diameter of thread, min. | Gage |
| 103. Major diameter of thread, min. | Gage |
| 104. Diameter of hole | Gage |
| 105. Thickness through body | Gage |
| 106. Thickness through disc | Gage |
| 107. Thread(s) damaged or not full | Visual |
| 108. Improper assembly | Visual |
| 109. Metal defective | Visual |
| 110. Protective coating, inadequate or damaged | Visual |
| **MINOR** | |
| 201. Pitch diameter of thread, max. | Gage |
| 202. Presence and location of wrench holes | Visual (see 6.1) |
| 203. Surface finish, improper | Visual (see 6.1) |
| 204. Radii or chamfers, missing or incorrect | Visual (see 6.1) |
| 205. Burr | Visual |
| 206. Foreign matter | Visual |

4.3.2.7 Carton Drawing B8597489.

CRITICAL

None defined

MAJOR

None defined



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4.3.2.7 Cont'd

| Defect | Method of Inspection |
|---|---|
| MINOR | |
| Prior to Sealing | |
| 201. Support missing or damaged | Visual |
| 202. Separators missing or damaged | Visual |
| 203. Improperly packed | |
| After Sealing | |
| 204. Contents loose | Manual |
| 205. Carton damaged, cut or gouged through all layers | Visual |
| 206. Carton not completely sealed | Visual |
| 207. Marking missing, incorrect or illegible | Visual |

4.3.2.8 Pallet, Drawing F7548604.

| | |
|---|---|
| CRITICAL | |
| None defined | |
| MAJOR | |
| None defined | |
| MINOR | |
| 201. Pallet damaged (unserviceable) | Visual |
| 202. Improperly packed | Visual |
| 203. Strapping missing, broken or loose | Visual and Manual |
| 204. Marking missing, incorrect or illegible | Visual |

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Mechanical properties.

4.3.3.1.1 Rotating band.- One body assembly shall be selected from each 8 hours production or fraction thereof from each banding machine in operation and subjected to this test to determine compliance with 3.4.2. Failure of the rotating band to comply shall be cause for rejection of the projectiles represented by the sample.

4.3.3.2 Band seating.



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4.3.3.2.1 Non-destructive method.- Ten body assemblies from each one hours production per banding machine in operation shall be subjected to this test. Failure of one or more bands to comply with 3.4.3 shall be cause for rejection of the assemblies represented by the sample.

4.3.3.2.2 Destructive method.- When the destructive method of testing is employed, one body assembly shall be selected from each 8 hours product on or fraction thereof per banding machine in operation. Failure of any band to comply with 3.4.3 shall be cause for rejection of the production represented by the sample.

4.3.3.3 Cleaning and application of protective coating.- The sample size and acceptance criteria for the inspection of cleaning the item and application of protective coating shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action recommended therein.

4.3.4 Functional.- The following Proving Ground tests will be performed by the Government.

4.3.4.1 First article sample.

4.3.4.1.1 Initial production sample.- The quantity listed in Table I shall be ballistically tested in the phase indicated.

Table I

| Phase | Quantity |
|---|---|
| Excess pressure | 10 |
| Service pressure | 10 |

4.3.4.2 Production lot sample.- The quantity listed in Table II shall be ballistically tested in the phase indicated.

Table II

| Phase | Quantity |
|---|---|
| Excess Pressure | 5 |
| Service Pressure | 5 |



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4.3.4.3 Excess pressure.- Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any projectile assembly to comply with 3.5 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.4.4 Service pressure.- Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any projectile assembly to comply with 3.5 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.5 Inspection equipment.- Index of Inspection Equipment Lists No. 10522528 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance wih the code designation on the applicable IEL.

4.4 Test methods and procedures.

4.4.1 Mechanical properties.

4.4.1.1 Rotating band.- The samples provided by 4.3.3.1.1 shall have the rotating band removed and carefully flattened without hammering so as to hold cold working to a minimum. A test specimen conforming to the largest obtainable "F" type specimen prescribed by Federal Test Method Standard No. 151 shall be removed from the flattened band, and tested in accordance with the procedures for this test contained in Federal Test Method Standard No. 151.

4.4.2 Band seating.

4.4.2.1 Tightness.

4.4.2.1.1 Nondestructive method.- The samples provided in 4.3.3.2.1 shall be tested utilizing the equipment and procedures prescribed in Frankford Arsenal Detroit Testing Machine Manual (see 6.2).

4.4.2.1.2 Destructive method.- The body assembly selected in accordance with 4.3.3.2.2 should not be processed through the band machining line, each band should have a light surface cut to true up the surface prior to measuring. The diameter of each band shall be measured on 3 locations, 60 degrees apart, on the points along the cylindrical portion of their length, near the front and rear edges. This comprises 6 diametrical measurements per band. These points of measurement shall be marked on the bands and on the projectile body adjacent thereto in such a manner as to permit proper body and band orientation after removal. The bands shall be carefully removed so as to keep deformation to a minimum, after removal, the thickness of the band shall be measured at each of the marked points (a total of 12 measurements) and the diameter of the band seat corresponding to these points shall also



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be measured. Add to each of the diameters of the band seat the sum of the corresponding 2 thicknesses of the band; the results obtained shall then be subtracted from the previously determined diameters over the band. The 6 results shall be totaled and the sum thus obtained will be divided by 6 to determine the average diametrical clearance.

4.4.3 Torque test.- In performing this test, the assembly torque prescribed by the applicable drawing shall be applied in the direction of assembly while the thread sealant is wet. Any perceptible movement of the component under test will constitute failure to comply with requirements.

4.4.4 Protective coating.- The procedures for testing the cleaning and surface treatment methods for the components and assemblies shall be in accordance with the criteria for testing as specified in the applicable specifications. The salt spray prescribed shall be performed in compliance with Federal Test Method Standard No. 151. A 20-percent sodium chloride solution shall be used.

4.4.5 Proving Ground Functional Tests.

4.4.5.1 Excess pressure.-The projectiles supplied in accordance with 4.3.4 for this phase will be assembled into complete rounds and fired for recovery in a gun for which the round is standard. A propellant charge weight adjusted to produce 115 percent of the service pressure of the round in a new gun shall be used. The projectile will be fired through a canvas or heavy muslin screen placed approximately 2000 meters from the muzzle of the gun. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile, for the initial production sample and the first 5 production lots. Prior to firing, paint will be applied to the windshield. Observation for paint on the cloth target will be made for windshield security.

4.4.5.2 Service pressure.- The projectiles supplied in accordance with 4.3.4 for this phase will be assembled into complete rounds utilizing the service charge and fired in a gun for which the round is standard for comparison with a like group from a reference lot. The sample projectile and reference projectile will be fired alternately through a canvas or heavy muslin screen placed approximately 2000 meters from the muzzle of the gun. Velocity and accuracy measurements will be obtained for this range accuracy in terms of maximum spread in inches, both horizontal and vertical and probable error (PE) in mils horizontal and vertical will be recorded for all rounds fired and forwarded to the cognizant technical agency. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile.



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5. PREPARATION FOR DELIVERY

5.1 Packing and Marking.

5.1.1 Level A - Not applicable.

5.1.2 Level B - Not applicable.

5.1.3 Level C - Packing and marking shall be in accordance with Drawing F-7548604.

5.1.3.1 Level C - Plug and Disc Assembly Drawing B8597489 shall be packed in accordance with MIL-P-10025. Pack 2000 plug and disc assemblies in 2 layers arranged in a 10 X 10 pattern, 10 to a cell, using interlocking separators.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.- When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Band seating.- The operating procedures for use of the Detroit Testing Machine as prepared by the Inspection Engineering Office, U. S. Army, Frankford Arsenal, may be used as an aid in checking band seating inspection procedures.

6.3 Ordering data.- Procurement documents should specify the title, number and date of this specification.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project no. 1315-A532



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